ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 2.01: Modular Arithmetic, Divisibility, Primes & Euler Totient

Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 8x + 28 in the domain of Modular Arithmetic, Primes & Divisibility. By completing the square f_1x = x - h^2 + k , find the sum k + f_16 .

[ ] A: 32
[ ] B: 28
[ ] C: 16
[ ] D: 26

Q2. A mathematical model in Modular Arithmetic, Primes & Divisibility is defined by the cubic function y_2x = 3x^3 - 4x^2 + 15x - 9 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 96
[ ] B: 57
[ ] C: 72
[ ] D: 34

Q3. In Modular Arithmetic, Primes & Divisibility, two displacement vectors are given by mathbf{u}_{3} = langle 4, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: -20
[ ] B: 80
[ ] C: 172
[ ] D: 60

Q4. The linear growth profile shown in the diagram for Modular Arithmetic, Primes & Divisibility follows an arithmetic progression with first term a_1 = 5 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 705
[ ] B: 89
[ ] C: 795
[ ] D: 1335

Q5. In Modular Arithmetic, Primes & Divisibility, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 16x,mathrm{d}x .

[ ] A: 104
[ ] B: 80
[ ] C: 96
[ ] D: 41

Q6. In Modular Arithmetic, Primes & Divisibility, consider the quadratic function f_6x = x^2 - 10x + 109 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_612 .

[ ] A: 222
[ ] B: 133
[ ] C: 217
[ ] D: 210

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Q7. The graph illustrates the cubic model y_7x = 3x^3 - 5x^2 + 33x - 16 studied in Modular Arithmetic, Primes & Divisibility. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 609
[ ] B: 496
[ ] C: 95
[ ] D: 529

Q8. In Modular Arithmetic, Primes & Divisibility, two displacement vectors are given by mathbf{u}_{8} = langle 5, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 79
[ ] B: 5
[ ] C: 74
[ ] D: 149